

Multiple Choice (10 marks)

1. If a random variable X is normally distributed with a mean of 118 and a standard deviation of 11, what Z-scores correspond to raw scores of 115, 134, and 99?
 - (a) -.27, 1.45, 1.73
 - (b) -.27, -1.45, -1.73
 - (c) -2.18, 0.15, -1.82
 - (d) 0, 1.5, -2
 - (e) .27, 1.45, 1.73
2. This biome contains plants whose roots cannot go deep due to the presence of a permafrost.
 - (a) Savanna
 - (b) Taiga
 - (c) Tundra
 - (d) Chaparral
 - (e) Alpine
3. What are the major forces of evolution?
 - (a) reproduction, migration, mutation, natural selection
 - (b) mutation, genetic drift, migration, genetic hitchhiking
 - (c) mutation, genetic drift, migration, random mating
 - (d) mutation, gene flow, migration, natural selection
 - (e) adaptation, genetic drift, migration, natural selection
4. Which group is composed entirely of individuals who maintained that species are fixed (i.e., unchanging)?
 - (a) Aristotle, Darwin, and Lamarck
 - (b) Aristotle, Lyell, and Darwin
 - (c) Aristotle, Linnaeus, and Cuvier
 - (d) Aristotle, Cuvier, and Lamarck
 - (e) Linnaeus, Darwin, and Lamarck
5. Fungi participate in each of the following EXCEPT
 - (a) photosynthesis to produce glucose

- (b) fermentation to produce alcohol
- (c) association with humans to produce ringworm
- (d) association with the roots of plants to form mycorrhizae

True / False (10 marks)

Answer True or False

6. True or False: Gene flow between populations increases the likelihood of speciation by enhancing genetic divergence.
7. True or False: Reducing the carrying capacity of the environment can effectively lower the K value and help control mosquito populations.
8. True or False: The extra Y chromosome is linked to an increase in artistic skills.
9. True or False: Commensalism is a form of interspecies interaction where one species benefits while the other is neither helped nor harmed.
10. True or False: Hydrolysis is involved in the conversion of starch to simple sugars.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the methods used to estimate the age of a rock, focusing on the role of radioactive elements and fossil content in determining geological timelines.
12. Discuss the concepts of incomplete dominance and epistasis in genetics, analyzing how they differ in terms of the interactions between alleles and genes.

End of Paper